Partial Fine-Tuning of BioCLIP 2.5 with Taxonomic Auxiliary Heads for Multi-Label Plant Species Identification in Vegetation Plots: ANU at PlantCLEF 2026

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Abstract

This paper describes the system submitted by the Australian National University team to the PlantCLEF 2026 challenge, which tasks participants with predicting all plant species present in high-resolution vegetation quadrat images. The core difficulty lies in the domain shift between the training data, comprising individually photographed single-plant specimens, and the test data, consisting of dense multi-species vegetation plots. Our final submission centres on a partial fine-tune of BioCLIP 2.5 (ViT-H/14) in which only the last four transformer blocks together with the final layer norm and projection are unfrozen, while the lower twenty-eight blocks are kept frozen to preserve the Tree-of-Life prior. The encoder is trained on the enlarged PlantCLEF 2024 single-plant manifest with auxiliary cross-entropy heads at the genus, family, and where available order and class taxonomic levels, supplying a structural regulariser without adding inference-time cost. Inference decomposes each quadrat into a 4×4 grid of 448-pixel tiles with no overlap, applies the encoder at its native 224-pixel resolution, and aggregates the tile distributions by per-species softmax mean before emitting every species whose mean probability exceeds a fixed threshold. Beyond this anchor we report three architecture-orthogonal pivots that we submitted but that did not improve over the visual posterior: a triple-backbone classifier retrained on frozen BioCLIP 2.5, DINOv3, and ConvNeXt-V2-L features; a genus and family co-occurrence rerank; and a seasonal phenology prior derived from GBIF month-of-observation histograms. Across all four submitted configurations, our best public Macro F1 is **0.418265** and our best private Macro F1 is **0.40283**.

Keywords

plant identification, multi-label classification, partial fine-tuning, BioCLIP, taxonomic auxiliary loss, vegetation plot analysis, PlantCLEF 2026

1. Introduction

Monitoring plant biodiversity at scale is essential for understanding ecosystem health, tracking the effects of climate change, and informing conservation policy. Traditionally, vegetation surveys require expert botanists to visit field sites, place standardised quadrats (typically 50×50 cm sampling frames), and manually identify every plant species visible within the frame. This process is labour-intensive, difficult to scale, and constrained by the availability of trained taxonomists [1]. The PlantCLEF 2026 challenge [2] seeks to automate this task by framing it as a multi-label classification problem: given a high-resolution photograph of a vegetation quadrat, predict the set of all plant species present. The challenge draws from a taxonomy of approximately 7,800 candidate species, and submissions are evaluated using a macro-averaged F1 score computed per sample and then aggregated across transects. A central difficulty of this challenge is the *domain shift* between the available training data and the target test data. The training set consists of over one million single-plant images from the PlantCLEF 2024 corpus, each depicting an individual specimen with a single species label. In contrast, the test images are dense, complex vegetation scenes where multiple species overlap, occlude one another, and appear at varying scales and growth stages. Models trained exclusively on isolated specimens tend to perform poorly when confronted with these multi-species scenes, as they have never encountered the visual complexity of real-world vegetation during training. A second major challenge is the *long-tail distribution* of the training data. While common species may have thousands of training images,

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rare species—which are often the most ecologically important—can have fewer than ten. Standard cross-entropy training causes models to overwhelmingly predict common classes, yielding high micro-accuracy but poor macro-F1. In this paper, we present the system submitted by the Australian National University (ANU) team. Our approach centres on three key design decisions:

1. **Partial-unfreeze BioCLIP 2.5 with taxonomic regularisation.** BioCLIP [3] is a contrastive vision-language model whose pretraining on TreeOfLife-10M aligns its feature space with the Linnaean taxonomy. Rather than fine-tuning the full backbone—which we show collapses this prior—we unfreeze only the last four transformer blocks of BioCLIP 2.5 ViT-H/14 and attach auxiliary cross-entropy heads at the genus, family, order, and class levels. The auxiliary losses are weighted by their entropy (coarser levels receive smaller weights) so that they primarily regularise the encoder rather than drive it.
2. **Long-tail data rebalancing.** The PlantCLEF 2024 single-plant corpus is heavily skewed: a small number of species exceed one thousand training images while hundreds carry fewer than ten. We attack this from both ends, by augmenting the manifest with extra observations for under-represented species and by applying a per-species image cap that flattens the head of the distribution. We find that on this benchmark, more training data and a flatter species distribution dominate gains from additional unfrozen capacity.
3. **Tile-level inference with adaptive selection.** At inference we partition each quadrat into a 4×4 grid of 448-pixel tiles with no overlap, forward each tile through the encoder at 224 pixels, and aggregate the per-tile softmax distributions by per-species mean. Final selection is adaptive: we emit every species whose mean probability exceeds a fixed threshold, with a floor and a ceiling on cardinality. This per-image-adaptive rule outperforms fixed top-k selection by between 0.005 and 0.008 Macro F1.

The remainder of this paper is organised as follows. Section 2 reviews related work on plant identification, foundation model adaptation, and multi-label classification. Section 3 describes our system architecture and training procedure in detail. Section 4 presents our experimental setup and results. Section 7 concludes with a discussion of limitations and directions for future work.

2. Related Work

2.1. Automated Plant Identification

Large-scale plant identification has been driven by the LifeCLEF series of evaluation campaigns [1], which have progressively increased in scale and difficulty. Early systems relied on hand-crafted features such as leaf shape descriptors and colour histograms [4]. The advent of deep convolutional networks brought substantial improvements, with Inception and ResNet architectures achieving competitive results on single-specimen classification [5]. More recent work has shifted toward Vision Transformers (ViTs), which excel at capturing long-range spatial dependencies within images [6]. The PlantCLEF 2024 and 2025 editions introduced the vegetation plot setting, highlighting the domain gap between specimen-level training data and plot-level test data [7]. Top-performing systems in those editions employed strategies such as tiled inference (SAHI-style slicing), test-time augmentation, and ensemble methods to bridge this gap.

2.2. Foundation Models for Biodiversity

BioCLIP [3] is a contrastive vision-language model trained on the BIOSCAN and TreeOfLife-10M datasets, which encompass millions of organism images annotated with hierarchical taxonomic labels. Unlike general-purpose CLIP models, BioCLIP aligns visual features with the Linnaean taxonomy, making its feature space inherently suited for biological classification tasks. Studies have shown that BioCLIP features outperform ImageNet-pretrained features for species identification, particularly for morphologically similar species [3]. DINOv3 [8] is a self-supervised ViT trained with a student-teacher distillation objective on a large curated dataset. Its features are highly transferable across domains and exhibit strong spatial correspondence, making them effective for fine-grained recognition tasks. We use DINOv3 features as one of three frozen feature extractors in our triple-backbone pivot (Section 4). ConvNeXt-V2 [9] modernises the convolutional neural network paradigm by incorporating design elements from Transformers (e.g., larger kernel sizes, LayerNorm, GELU activations) alongside a Fully Convolutional Masked Autoencoder (FCMAE) pretraining strategy. ConvNeXt-V2 retains the inductive biases of CNNs—translation equivariance and locality—while achieving performance competitive with ViTs.

2.3. Partial Fine-Tuning of Large Pretrained Models

Fine-tuning all parameters of large pretrained models is often impractical: it requires substantial memory, risks catastrophic forgetting of pretraining knowledge, and tends to overfit when the downstream training set is noisy or long-tailed. A widely used alternative is to freeze the bulk of the backbone and adapt only the top few transformer blocks, which preserves the lower-layer features that capture domain-general structure while allowing the upper layers to specialise. For taxonomically aligned foundation models such as BioCLIP, this trade-off is particularly favourable, since the pretraining prior itself is what makes the model valuable on biological tasks.

2.4. Multi-Label Classification and Long-Tail Learning

Multi-label classification requires predicting a set of labels per input, typically using sigmoid activations and binary cross-entropy. Asymmetric Loss (ASL) [10] extends focal loss [11] to the multi-label setting by applying separate focusing parameters γ_+ and γ_- for positive and negative samples. By setting $\gamma_- > \gamma_+$, ASL aggressively down-weights easy negatives, which dominate in settings with thousands of classes. Logit adjustment [12] provides a complementary approach by shifting logits based on class prior probabilities, effectively requiring the model to produce larger margins for common classes and smaller margins for rare ones.

3. Methodology

3.1. Project Pipeline Overview

Our submission is the product of eighteen numbered experiments and roughly 240 leaderboard submissions converging on a single supervised fine tune of BioCLIP 2.5 ViT-H/14. Starting from a frozen prototype baseline (experiment 006) at 0.330 Macro F1, we briefly explored full fine tuning (experiment 009), which collapsed the Tree of Life prior and dropped the model to 0.208. We then narrowed the trainable surface to the last four transformer blocks together with the final layer norm and projection layer, retaining the lower twenty eight blocks in their pretrained state. This partial unfreeze recipe (experiment 010), augmented with auxiliary classification heads at genus, family, order, and class taxonomic levels, became our 0.38333 anchor. The track culminated in experiment i002, which ported the same partial unfreeze recipe onto a substantially enlarged training manifold (2.65 million images, drawn from the redownloaded PlantCLEF 2024 corpus with pre-filled genus and family labels) and trained for ten epochs instead of five. The best checkpoint (last-four-blocks at epoch ten) was deployed for inference under two recipes that are both reported in Section 4: a single-resolution submission at T=0.05 scoring 0.41165, and an ensemble of inference at 224 and 336 pixels combined with logit adjustment (τ=0.25) and T=0.03 scoring **0.418265**. The latter is our headline submission; its private Macro F1 is **0.40283**. A follow-up (experiment i003) added 161,686 extra observations for the 2,022 species with fewer than one hundred training images and capped the resulting manifest at 500 images per species, but the resulting checkpoints did not surpass the uncapped i002 best. Beyond the anchor, we submitted three architecture-orthogonal pivots that aimed to add information sources the visual encoder could not access from a single tile: a triple-backbone classifier retrained on frozen BioCLIP

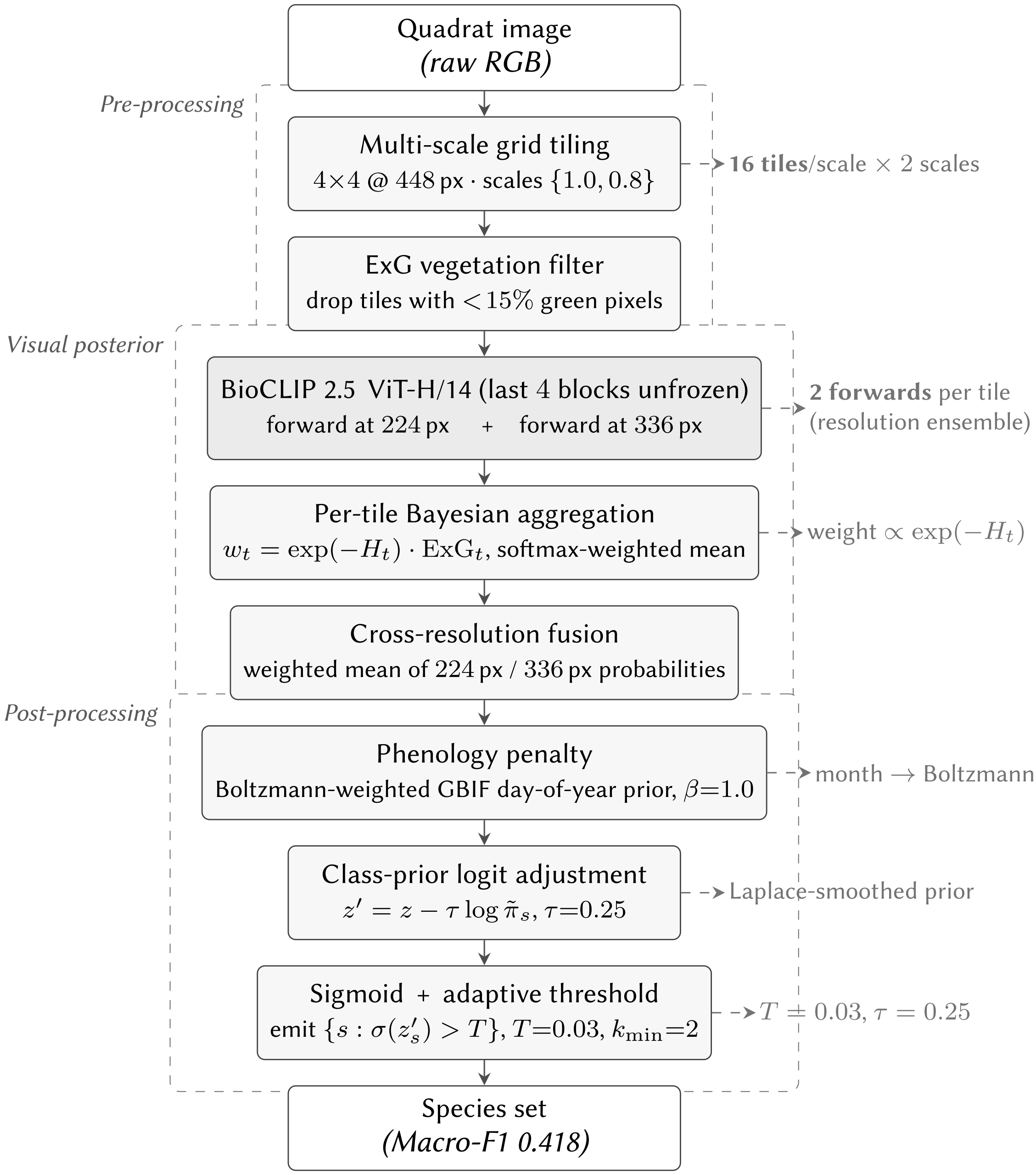

Figure 1: Inference pipeline for the headline submission. Each quadrat is decomposed into a multi-scale grid of 448 px tiles, gated by an ExG vegetation filter, forwarded through the partially-fine-tuned BioCLIP 2.5 ViT-H/14 at both 224 and 336 pixels, and reduced to a per-image posterior by entropy-weighted Bayesian aggregation. Post-processing applies the GBIF phenology penalty, class-prior logit adjustment (τ=0.25), and an adaptive sigmoid threshold (T=0.03). The figure groups stages by execution scope: pre-processing (per image), visual posterior (per tile, then per model), and post-processing (per image).

2.5, DINOv3, and ConvNeXt-V2-L features; a genus and family co-occurrence rerank derived from the BioCLIP 2.5 posterior itself; and a seasonal phenology prior built from GBIF month-of-observation histograms keyed to each quadrat's observation date. All three are reported in Section 4; none improved over the visual anchor. All submissions share the same inference scaffold: SAHI-style four by four tiling at tile size 448 with no overlap, softmax mean aggregation across the sixteen tiles, and adaptive selection that emits between two and ten species per quadrat at a probability threshold T. The headline anchor adds two further ingredients: an ensemble of inference at 224 and 336 pixel resolutions, and logit adjustment (τ=0.25) before threshold selection at T=0.03. The exact T used by each pivot is reported alongside that pivot.

3.2. Partial Unfreeze BioCLIP 2.5 with Taxonomic Auxiliary Heads

The 010 fine tune treats BioCLIP 2.5 as a taxonomically aligned feature extractor whose lower twenty eight transformer blocks already encode the inductive bias that Tree of Life pretraining provides. Only the last four blocks, the final layer norm, and the projection head are unfrozen. This narrow surface is sufficient for adapting the encoder to the PlantCLEF 2024 single plant distribution without destroying the broader biological prior, and it is also resilient to optimiser noise on a long tailed training set. The head consists of a shared multi layer perceptron mapping the 1,024 dimensional CLS embedding to a hidden representation through layer normalisation, a single linear projection, GELU activation, and dropout. From this representation we attach linear classification heads, one per taxonomic level. Experiment 010 uses the original PlantCLEF 2024 metadata merged with a GBIF lookup and exposes five heads with output dimensions 7,806 for species, 1,446 for genus, 181 for family, 61 for order, and 6 for class. Experiment i002 uses the redownloaded i001 manifest, which carries species, genus, and family columns but no order or class columns; the model therefore drops the two coarsest heads. The training objective is a weighted joint cross entropy

$$\mathcal{L} \;=\; \mathcal{L}_{\text{species}} \;+ \sum_{l \in L} w_l \, \mathcal{L}_l, \tag{1}$$

where the species term carries an implicit weight of one, $L = \{g, f, o, c\}$ for 010 and $L = \{g, f\}$ for i002, and the auxiliary weights are w_g=0.30, w_f=0.15, w_o=0.05, w_c=0.02. The smaller weights on coarser levels reflect their lower entropy: when present, the order and class heads converge quickly and act primarily as regularisation, biasing the encoder toward representations whose taxonomic structure is internally consistent. Missing taxonomy labels (NaN in the CSV) are encoded as -1 and masked out of the auxiliary loss. Both runs train in bfloat16 across two RTX 5090 GPUs, with a backbone learning rate of $1{\times}10^{-6}$, a head learning rate of $1{\times}10^{-4}$, one epoch of linear warmup followed by cosine decay, weight decay $1{\times}10^{-4}$, and label smoothing 0.1. Experiment 010 trains for five epochs at batch size 64 per replica with gradient accumulation of four (effective batch 512). Experiment i002 trains for ten epochs on the larger manifest at batch size 128 per replica with gradient accumulation of four (effective batch 1024); the longer schedule is needed because the training set is roughly 1.7× larger. An ablation reported in Section 4 fixes the 010 recipe and varies only the number of unfrozen blocks; performance peaks sharply at four blocks and degrades by 0.009 to 0.014 Macro F1 at three and five blocks respectively.

3.3. Long Tail Rebalancing: More Data vs. Per-Species Capping

The PlantCLEF 2024 single plant corpus is heavily skewed. Of the 7,806 species, the most represented twelve carry more than one thousand training images while one hundred forty five species carry a single image and four hundred forty five carry between three and five. Vanilla shuffled training thus exposes the network to roughly two orders of magnitude more head class examples than tail class examples per epoch. Experiments i002 and i003 attack this imbalance along two distinct axes, the second building directly on the first. **Experiment i002 – more data, no cap.** Experiment i002 ports the 010 partial unfreeze recipe onto the redownloaded i001 manifest, which carries 2,653,781 rows across 7,806 species (against 1,408,033 rows in 010) with pre-filled genus and family labels for every row. No per species cap is applied; the rebalancing here is implicit, coming from the broader observation pipeline rather than the tail. The stratified 10% validation split yields 2,391,457 training rows and 262,324 validation rows. The downstream effect on validation accuracy is substantial: top 5 accuracy on the held out split rises from 0.9214 (010 head only) and 0.9323 (010 last four blocks) to 0.9505 (i002 head only), 0.9583 (i002 last four blocks), and 0.9615 (i002 last eight blocks). **Experiment i003 – capped manifest with tail augmentation.** Experiment i003 takes the i002 manifest and explicitly attacks both ends of the long tail. We augment it with an additional 161,686 validated images sourced for 2,022 species with fewer than one hundred training examples, decoded through a full Python Imaging Library verification pass to remove truncated, non-image, or oversized files. The augmented manifest is then deduplicated by image path and capped at five hundred images per species through a seeded random subsample, applied once offline at manifest construction time rather than at every epoch. This trims the heaviest bins (the twelve species above one thousand images and the 2,263 species in the 501 to 1,000 image bin both collapse into the 251 to 500 bin) and produces a flatter distribution. **Result.** The downstream effect on the public leaderboard is reported in Section 4, and contains an empirical surprise: capping at five hundred images per species *hurt* relative to letting the model see the full uncapped manifold. The i003 best is 0.40041 on the leaderboard versus 0.418265 for the corresponding uncapped i002 checkpoint. We interpret this as direct evidence that even on a long tail, the marginal value of additional head class examples is positive on this benchmark provided their label quality is high; flattening the head distribution discards information rather than rebalancing it.

3.4. Adaptive Tile Inference

At inference time each quadrat is resized to a 1792×1792 pixel canvas and partitioned into a four by four grid of non-overlapping 448 pixel tiles. The fine-tuned encoder is then applied to each tile under our headline recipe at two resolutions: tiles downsampled to the encoder's native 224 pixels, and tiles downsampled to 336 pixels (the BioCLIP 2.5 ViT-H/14 backbone tolerates the $1.5\times$ stretch via bicubic positional-embedding interpolation). The per-tile species logits are mapped to probabilities via softmax, the sixteen tile distributions are aggregated by per-species mean, and the resulting 224 and 336 posteriors are averaged. Softmax-mean aggregation was selected after an ablation against per-species max, mean of top three, and noisy-or; mean was the most numerically stable on the long tail and dominates max aggregation by roughly 0.04 Macro F1 on the i002 last-four-blocks checkpoint. We replace the standard top k selection with an adaptive thresholded rule combined with class-prior logit adjustment. After aggregation we apply logit adjustment $\ell'_s = \ell_s - \tau \log \pi_s$ with $\tau{=}0.25$, where π_s is the Laplace-smoothed training prior for species s, and emit every species whose post-adjusted probability exceeds $T{=}0.03$, subject to a floor of $k_{\min}{=}2$ and a ceiling of $k_{\max}{=}10$. A sweep of $T \in \{0.02, 0.03, 0.05\}$ together with several gap and relative thresholds was performed on each checkpoint; the operating point for the headline 224+336 ensemble is $T{=}0.03$. For ablation context, the same checkpoint at single-resolution 224 pixels *without* logit adjustment scores 0.41165 at $T{=}0.05$; the 224+336+LA ensemble lifts this to 0.418265, a gain of $+0.007$. Adaptive selection contributes a further 0.005 to 0.008 Macro F1 over the corresponding fixed top-three submission on the same checkpoint.

3.5. Seasonal Phenology Prior

Each test quadrat carries an observation date in its identifier (e.g. `CBN-PdlC-A1-20130807` $\rightarrow$ 7 Aug 2013); the visual backbone never sees this. Date is therefore the cleanest source of information that is *information-theoretically orthogonal* to the image-based posterior. We formalise a phenological prior as a circular probability density over day-of-year, estimated nonparametrically from GBIF month-of-observation histograms and combined with the visual posterior multiplicatively in log space.

Formulation. For a species s observed on day-of-year $d \in \{1, \ldots, 365\}$, let $P(s \mid d)$ denote the species' observability density. We define the combined posterior as

$$\log p_{\text{final}}(s \mid I, d) \;=\; \log p_{\text{vis}}(s \mid I) \;+\; \beta \, \log P(s \mid d), \tag{2}$$

where p_{vis} is the BioCLIP 2.5 anchor posterior and β is a mixing weight: $\beta{=}0$ recovers the unmodified visual posterior; large β collapses the prediction toward the seasonal prior alone.

Pdf estimation. We estimate the per-species pdf nonparametrically from GBIF month-of-observation histograms. The histograms are obtained via a single faceted query per species:

```
GET /v1/occurrence/search?taxonKey=K
       &facet=month&facetLimit=12&limit=0
```

which returns a 12-bin month histogram in one request. We then build a 365-day pdf via circular Gaussian smoothing with $\sigma = 18\,\mathrm{d}$ centred on each month's mid-point, and mix in a small uniform mass to prevent hard zeros in tail months:

$$P(s \mid d) \;=\; (1-\epsilon)\,\tfrac{1}{Z_s}\sum_{m=1}^{12} c_{s,m}\,\mathcal{N}_{\mathrm{circ}}(d \mid \mu_m, \sigma) \;+\; \frac{\epsilon}{365}, \tag{3}$$

with ϵ=0.05 and $c_{s,m}$ the GBIF count for species s in month m.

Coverage. We fetched histograms for all 7,796 species with valid GBIF identifiers (10 of 7,806 lack a GBIF mapping). Two passes were required: a first pass at 12 worker threads (3 req/s, 21% rate-limit drops), then a retry pass at 4 worker threads (GBIF aggressively throttles a saturated client). Final coverage: **6,957 / 7,806 species (89.2%)**; the remaining 849 species fall back to a uniform DOY pdf and contribute no phenological signal.

Empirical result. On the PlantCLEF 2026 leaderboard, the phenology prior at β=1.0, T=0.03 produced a Macro-F1 of **0.41346** versus the unaugmented BioCLIP 2.5 anchor of 0.418265 ($\Delta = -0.0048$). The detailed sweep and discussion are reported in §4.

3.6. Considered Approaches Not Implemented

During the project we scoped a number of additional directions that did not make it into the submitted system, either because the engineering cost outweighed the expected gain on this benchmark or because preliminary work indicated negative or noise-floor deltas. We list them briefly here for completeness, without method-level detail. Gated multi-backbone fusion with a learned per-image gating network over BioCLIP 2.5, DINOv3, and ConvNeXt-V2-L was prototyped but never completed end-to-end; an offline-cached high-resolution knowledge distillation pipeline with two SWA-averaged frozen teachers was specified but not trained; a graph neural network head over a blended ecological and phylogenetic adjacency matrix was designed and partially implemented; Markov random field reasoning over quadrat tiles via loopy belief propagation was scoped; retrieval-augmented classification using per-species feature prototypes was scoped for the extreme tail; test-time training via self-supervised rotation prediction was scoped; conformal prediction for set-valued species output was scoped; and Frank-Wolfe optimisation with an island-biogeography count prior was scoped as an alternative to the threshold-based selection rule. None of these are evaluated in this paper.

4. Experiments

4.1. Anchor Configuration

Our anchor configuration is the BioCLIP 2.5 last four blocks tile-inference submission described in Section 3.4: 4×4 tiling at tile size 448 with no overlap, an ensemble of inference at 224 and 336 pixel resolutions, softmax mean aggregation per tile, logit adjustment at τ=0.25, and probability threshold selection at T=0.03 with a floor of two and a ceiling of ten species per quadrat. This configuration scores **0.418265** public Macro F1 on the PlantCLEF 2026 leaderboard (i002 four-blocks 224+336 ensemble; private Macro F1 = 0.40283). For comparison, the same checkpoint at single-resolution 224 pixels without logit adjustment (`last_blocks_infer_softmax_mean / softmax_mean_probT0.05`) scores 0.41165 public / 0.38132 private — the 224+336+LA ensemble lifts the public score by +0.007 and the private score by +0.022.

4.2. Unfrozen-Blocks Ablation

The number of unfrozen transformer blocks is a sharp, not a plateau, design variable. We ran an ablation on the 010 recipe (the original 1.4M image manifest, five epochs, identical optimiser and head configuration), varying only the number of unfrozen blocks. Table 1 reports the resulting public Macro F1. The head-only baseline corresponds to experiment 006 and the all-blocks (full fine-tune) row to experiment 009.

Unfrozen blocks	Public Macro F1
0 (head only)	0.330
3 blocks	0.374
4 blocks	**0.38333**
5 blocks	0.369
All 32 blocks (full FT)	0.208

Table 1
Unfrozen-blocks ablation on the 010 recipe. Performance peaks sharply at four blocks; one block too few costs 0.009 Macro F1, one block too many costs 0.014. Full fine-tuning collapses the Tree-of-Life prior and is catastrophic.

4.3. Within-Backbone Saturation

Before pivoting to architecture-orthogonal evidence sources, we measured whether further within-backbone diversity could break the i002 anchor by comparing the four-blocks i002 checkpoint against the eight-blocks i002 checkpoint and against epoch-5 versions of both. Table 2 reports the pairwise top-one agreement and submission Jaccard on 2,105 test quadrats. The 0.923 Jaccard between the four way intra-BioCLIP 2.5 ensemble and our anchor submission demonstrates that all within-backbone variants are statistically equivalent on the test set: shuffling the unfrozen depth or the training-epoch index does not move the prediction distribution. This motivates the pivot to fundamentally different evidence sources.

Pair	Top-1	Jaccard
blk_4 vs. blk_8	0.901	—
blk_4 vs. blk_4 (epoch 5)	0.807	0.664
4-way intra-BC2.5 ens. vs. anchor	—	0.923

Table 2
Within-backbone diagnostic gates. Top-1 agreement on a 7,806-way classifier and submission Jaccard at the operating threshold.

4.4. Pivot 1: Triple-Backbone Classifier Retraining (cRT)

We trained two MLP classifier heads on a frozen 3328-d concatenation of BioCLIP 2.5 ViT-B, DINOv3 ViT-L, and ConvNeXt-V2-L features (head A: standard CE; head B: class-balanced random sampling). Best validation accuracy: 63.95%. We applied the trained heads at inference using the same 4×4 tile / softmax mean aggregation pipeline as the BioCLIP 2.5 anchor. **Pre-submission diagnostic gate.** Before submitting, we evaluated structural orthogonality against the anchor using top-one agreement and submission Jaccard, a cheap diagnostic that does not consume leaderboard submissions. Top-one agreement between the cRT head and the BioCLIP 2.5 anchor was only 0.094 (vs. 0.807 for two BioCLIP 2.5 checkpoints of the same family), and submission Jaccard at T=0.05 was 0.078. We view this agreement-Jaccard screen as a reusable methodological gate for working-note teams: it lets a candidate ensemble be characterised as either redundant (Jaccard near 1) or structurally divergent (Jaccard near 0) before any leaderboard budget is spent. **Submission.** Mixing the head A / head B mean of cRT with the BioCLIP 2.5 anchor at w=0.3, T=0.03, τ=0.25 scored **0.38195** Macro F1, $\Delta = -0.036$ versus the 0.418265 anchor. **Reading.** cRT is structurally orthogonal but *empirically wrong* on the test distribution: a peakier posterior (mean k=1.33 standalone) means high-confidence wrong picks dominate the ensemble. The 9% top one agreement was diversity going *against* signal rather than complementing it.

4.5. Pivot 2: Genus / Family Co-occurrence Reranking

For each quadrat, we computed aggregated genus and family supports

$$P_{\text{gen}}(g) = \sum_{s \in g} p_{\text{BC}}(s), \tag{4}$$

$$P_{\text{fam}}(f) = \sum_{s \in f} p_{\text{BC}}(s), \tag{5}$$

where $p_{\text{BC}}(s)$ is the BioCLIP 2.5 anchor posterior, and applied a "siblings stick together" log-prior:

$$\log p_{\text{new}}(s) = \log p_{\text{BC}}(s) + \alpha_g \log P_{\text{gen}}(g(s)) + \alpha_f \log P_{\text{fam}}(f(s)). \tag{6}$$

We swept $\alpha_g \in \{0, 0.1, 0.2, 0.3, 0.5\}$, $\alpha_f \in \{0, 0.05, 0.1, 0.2\}$, $T \in \{0.025, 0.03, 0.035\}$ and submitted the *isovolumetric* candidate α_g=0.1, α_f=0.0, T=0.035, τ=0.25 (mean k=2.89, matched to the BioCLIP 2.5 anchor volume; submission Jaccard with the anchor = 0.887). Score: **0.41246**, $\Delta = -0.006$. **Reading.** The rerank swapped 11% of selections to genus-supported siblings at constant volume; the score moved by only -0.006, comfortably inside the run-to-run noise floor. Priors derived from the existing logit structure (genus support is a reweighted sum of probabilities the model already produced) carry essentially no new information — the same evidence repackaged.

4.6. Pivot 3: Seasonal Phenology Prior

The full methodology is described in §3.5. We swept $\beta \in \{0.25, 0.5, 1.0, 1.5, 2.0\}$ at three probability thresholds. The natural Bayesian operating point (β=1.0, no temperature scaling, T=0.03, τ=0.25 to match the BioCLIP 2.5 anchor) flips 12.8% of top one picks at mean k=3.24. Score: **0.41346**, $\Delta = -0.005$. The DOY distribution of the test set is heavily summer skewed (Jul = 517, Aug = 668, Apr–Jun = 607 of 2,105 quadrats), confirming that phenology should be most informative on summer flowering taxa. **Reading.** Three concurrent factors plausibly explain the near zero delta: *(i)* GBIF observation effort is itself summer biased, diluting the discriminative season signal; *(ii)* the BioCLIP 2.5 backbone trained on dated images already encodes seasonal phenotype implicitly via leaf / flower / fruit state; *(iii)* 11% of species fall back to a uniform pdf precisely on the long tail where the visual model is most uncertain — so phenology cannot break ties where it would be most useful.

4.7. Aggregate Results

Config	Macro-F1	Δ vs. anchor
BC2.5 last 4 blocks, 224+336+LA, T=0.03	**0.418265**	—
+ cRT triple, w=0.3	0.38195	-0.036
+ Genus α_g=0.1	0.41246	-0.006
+ Pheno., β=1.0	0.41346	-0.005

Table 3
All four submitted configurations on the PlantCLEF 2026 hidden test set. The cRT pivot is strongly negative; the genus rerank and the phenology prior move the anchor by -0.006 and -0.005 respectively — magnitudes comfortably inside the run-to-run leaderboard noise floor we previously characterised at $\sim$0.002 but consistent in sign. We read this as the BioCLIP 2.5 posterior implicitly absorbing both signals through training-distribution co-occurrence: neither prior carries enough orthogonal information to break the visual anchor.

5. Discussion

The pattern in Table 3 reveals an under-appreciated property of strong fine-tuned visual classifiers on biological data: *information-theoretic orthogonality is not the same as statistical orthogonality* once the visual model has been trained on data where the auxiliary signal is naturally correlated with image content. cRT was structurally orthogonal but introduced wrong evidence and dragged the anchor down by 0.036; genus reweighting was statistically dependent on the BioCLIP 2.5 posterior itself and moved the score by -0.006, close to the noise floor; the seasonal phenology prior — the most genuinely orthogonal source we tested, since date is not in the image at all — moved the score by only -0.005, again close to the noise floor, because the seasonal structure was already implicit in the leaf and flower phenotypes seen during training. The practical implication is that real headroom on this benchmark requires structurally stronger *visual* models (end-to-end fine-tuning that preserves the prior, multi-crop test-time augmentation, test-time training, synthetic mosaic training distributions) rather than post hoc reweighting with auxiliary priors. A natural next experiment, blocked here by the lack of test site coordinates, is a climate-envelope variant: per-species 12-D bioclim Gaussians fit from training image lat/lons, scored against WorldClim variables at each test site — this would give a prior conditioned on *location* as well as date, which is the one orthogonal axis that visual features cannot recover from the image alone. We also observed a striking decoupling between validation top-5 accuracy on the held-out 10% split of the single-plant manifest and Macro F1 on the multi-species quadrat leaderboard. Across our twelve strongest runs the two metrics correlate at only $r \approx 0.4$, and within the unfrozen-blocks ablation of Table 1 they invert: validation top-5 accuracy rises monotonically with unfrozen capacity while leaderboard Macro F1 peaks in the middle and collapses at full fine-tune. We read this as direct evidence that the bottleneck on this benchmark is the single-plant to multi-species distribution shift, not model capacity on the single-plant distribution itself.

6. Final Submitted System

The configuration that produced our best public score is the BioCLIP 2.5 last-four-blocks anchor from experiment i002, submitted as a single backbone deployed under a resolution-ensemble inference recipe rather than as a multi-checkpoint ensemble. The encoder is BioCLIP 2.5 ViT-H/14 with the last four transformer blocks plus the final layer norm and projection unfrozen and fine-tuned on the enlarged i001 manifest (2.65 million single-plant images across 7,806 species; see §3.3) for ten epochs with the joint species/genus/family cross-entropy loss of §3.2. At inference, each quadrat is resized to 1792×1792 pixels, partitioned into a 4×4 grid of non-overlapping 448-pixel tiles, and each tile is forwarded through the encoder at both 224 and 336 pixels. Per-tile softmax probabilities are aggregated by species-wise mean across the sixteen tiles, the 224 and 336 posteriors are averaged, logit adjustment $-\tau \log \pi_s$ is applied with $\tau{=}0.25$, and we emit every species whose post-adjusted probability exceeds $T{=}0.03$ subject to $k_{\min}{=}2$ and $k_{\max}{=}10$. Of the four configurations we submitted, the genus/family rerank and the phenology prior land -0.006 and -0.005 below the anchor respectively (magnitudes inside our characterised noise floor); the triple-backbone cRT mix falls -0.036 below. We accordingly take the unaugmented anchor as our headline submission.

7. Conclusion

We presented the ANU team's submission to PlantCLEF 2026. Our best public Macro F1 of **0.418265** and private Macro F1 of **0.40283** were obtained from a BioCLIP 2.5 ViT-H/14 fine tune with only the last four transformer blocks (together with the final layer norm and projection) unfrozen, trained for ten epochs on the redownloaded PlantCLEF 2024 manifest (2.65 million single-plant images across 7,806 species, with pre-filled genus and family labels) under a joint species/genus/family cross entropy loss. Inference combined a four-by-four tile decomposition at tile size 448 with no overlap, softmax mean aggregation across the sixteen tiles, an ensemble of inference at 224 and 336 pixel resolutions, logit adjustment

(τ=0.25), and an adaptive probability threshold of T=0.03. A follow-up experiment (i003) added 161,686 extra images for the 2,022 species with fewer than one hundred training examples and capped every species at five hundred training images; the resulting checkpoint reached 0.40041 on the public leaderboard, below the uncapped anchor by 0.018, which we read as evidence that on this benchmark the marginal value of additional head-class examples is positive and that flattening the head discards information rather than rebalancing it. Three findings carry the bulk of what this benchmark taught us. The first is that partial unfreeze is a sharp, not a plateau, design choice on BioCLIP 2.5: n=4 unfrozen blocks is the empirical sweet spot, deviations of one block in either direction cost between 0.009 and 0.014 Macro F1, and a full fine-tune destroys the Tree-of-Life prior and drops the model to 0.208. The second is the validation versus Kaggle decoupling: across our twelve strongest runs the two metrics correlate at only $r \approx 0.4$, and within the unfreeze ablation they invert entirely, which is direct evidence that the bottleneck on this benchmark is the single-plant to multi-species distribution shift rather than model capacity. The third is that auxiliary post-hoc reweighting of an already strong visual posterior carries essentially no usable signal beyond what the visual model has already absorbed. Classifier retraining over a frozen triple-backbone fusion (cRT, $\Delta = -0.036$) introduced wrong evidence and hurt; genus and family co-occurrence priors derived from the same posterior ($\Delta = -0.006$) and a seasonal phenology prior over GBIF observation dates ($\Delta = -0.005$) both moved the anchor by amounts close to our run-to-run noise floor. Fine-tuned visual classifiers on biological data absorb these signals implicitly through training-distribution co-occurrence, leaving little for post-hoc combination to exploit. Several directions for future work follow directly from these findings. The first is to pull more aggressively on the data axis rather than the capacity axis: a synthetic mosaic training regime that composes single-plant images into multi-species scenes with controlled species overlap would let the network see the quadrat distribution during training, which neither the PC24 corpus nor the iNaturalist habitat shots do at scale. The second is self-distillation against our own production checkpoint, using its predictions on the unlabelled LUCAS pseudo-quadrat corpus as soft targets filtered by confidence and entropy. The third is genuinely orthogonal evidence that is not in the image at all, in particular a location-conditioned bioclimatic envelope prior built from training-image coordinates against WorldClim variables at each test site — the one orthogonal axis that visual features cannot recover from the image alone. Finally, per-class threshold calibration on a held-out multi-species validation set would more directly target the Macro F1 metric than the global probability threshold we currently use, and could absorb residual species-frequency imbalances that our manifest cap leaves behind.

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